

Database Management Systems (DBMS)

Comprehensive Exam Notes & Relational Mapping Guide

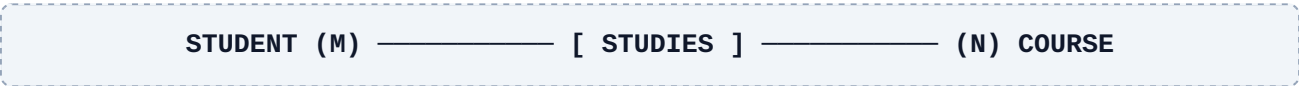
TOPIC: MANY-TO-MANY (M:N)

EXAM PRIORITY: HIGH

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1 What is a Many-to-Many Relationship?

A **Many-to-Many (M:N) relationship** exists when multiple entities from one entity set can be associated with multiple entities from another entity set.



Core Concept:

- One **Student** can enroll in / study multiple courses.
- One **Course** can be taken by / studied by multiple students.

Concrete Mapping Example:

- Student 1 → Course C1
 - Student 1 → Course C2
 - Student 2 → Course C1
- Student 2 → Course C3
 - Student 3 → Course C1

Notice that Student 1 studies multiple courses (C1, C2), and Course C1 is studied by multiple students (1, 2, 3). Hence, it is an M:N relationship.

2 Converting M:N ER Model into Relational Model

When mapping an ER diagram containing an M:N relationship to relational tables, we must convert both the entity sets and the relationship set into distinct tables.

1. Student Table

Roll_No (PK)	Name	Age
1	A	20
2	B	21
3	C	22
4	D	20

2. Course Table

Course_ID (PK)	Course_Name	Credits
C1	DBMS	4
C2	Networks	3
C3	Java	4
C4	Python	3

3. Relationship Table (Studies)

Roll_No	Course_ID
1	C1
1	C2
2	C1
2	C3
3	C1
3	C4

Why is the separate Relationship Table necessary?

The Student table only maintains student details, and the Course table only maintains course details. Neither table individually can maintain which student studies which course without introducing extreme data redundancy. The **Studies** relationship table bridges this gap.

3 Keys in the Relationship Table ★★★★★

Foreign Key References:

The Studies table contains two primary attributes referencing the base entities:

- **Roll_No** → Foreign Key referencing Student (Roll_No)
- **Course_ID** → Foreign Key referencing Course (Course_ID)

Student(Roll_No) ← [FK] — Studies(Roll_No, Course_ID) — [FK] → Course(Course_ID)

Determining the Primary Key:

✗ Can Roll_No alone be the Primary Key?

No. Roll_No repeats (e.g., Roll_No 1 appears for C1 and C2). Primary keys cannot contain duplicate values.

✗ Can Course_ID alone be the Primary Key?

No. Course_ID repeats (e.g., C1 appears for Roll_No 1, 2, and 3).

✓ Solution: Composite Primary Key

By combining both attributes → **(Roll_No, Course_ID)**, each tuple becomes completely unique.

Primary Key of Studies Table: Composite Primary Key (Roll_No, Course_ID)

4 Table Reduction & Merge Analysis

Can we reduce the number of tables in an M:N relationship?

Answer: **NO ✗** (Minimum 3 tables required).

Why can't we merge Student and Studies?

If we attempt to combine Student + Studies into a single table:

Roll_No	Name	Age	Course_ID
1	A	20	C1
1	A	20	C2
2	B	21	C1

Result: Student attributes (Name, Age) are unnecessarily duplicated for every course enrolled. This leads to insertion, deletion, and update anomalies.

Comparison Across Relationship Types (Exam Shortcut):

Relationship Type	Primary Key of Relationship Table	Can Tables Be Reduced?	Final Table Count
1 : 1 (One-to-One)	Primary Key of <i>either side</i>	✓ Yes	2 or 1
1 : N (One-to-Many)	Primary Key of the <i>MANY side</i>	✓ Yes (Merge with Many side)	2

Relationship Type	Primary Key of Relationship Table	Can Tables Be Reduced?	Final Table Count
M : N (Many-to-Many)	Composite Key (Combination of both)	✗ No	3

5 The Golden Rules & Exam Summary

★ GOLDEN RULES FOR ER-TO-RELATIONAL MAPPING

1 : 1 Relationship: Either key can identify the relationship → Tables can be reduced by merging.

1 : N Relationship: Key from the MANY side becomes the relationship Primary Key → Merge relationship table into the MANY side entity.

M : N Relationship: Both primary keys are required to form a **Composite Primary Key** → Reduction NOT possible (Requires 3 separate tables).

🔥 One-Line Memory Trick for Exams:

“ 1:1 → Either | 1:N → Many | M:N → Both ”

🎯 Quick Checklist for Final Revision:

1. M:N Relationship always results in **3 tables** in the Relational Schema.
2. Relationship table contains **2 Foreign Keys** referencing parent tables.
3. Primary Key of the relationship table is a **Composite Primary Key** composed of both foreign keys.
4. Table count **cannot** be reduced without creating redundancy or anomalies.